OBLIQUE SPHERICAL TRIANGLE

SPHERICAL TRIGONOMETRY

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

70%

WEEKLY

30%

EXAM

WEEKLY GRADE

70%

LONG QUIZ

30%

SEATWORK

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

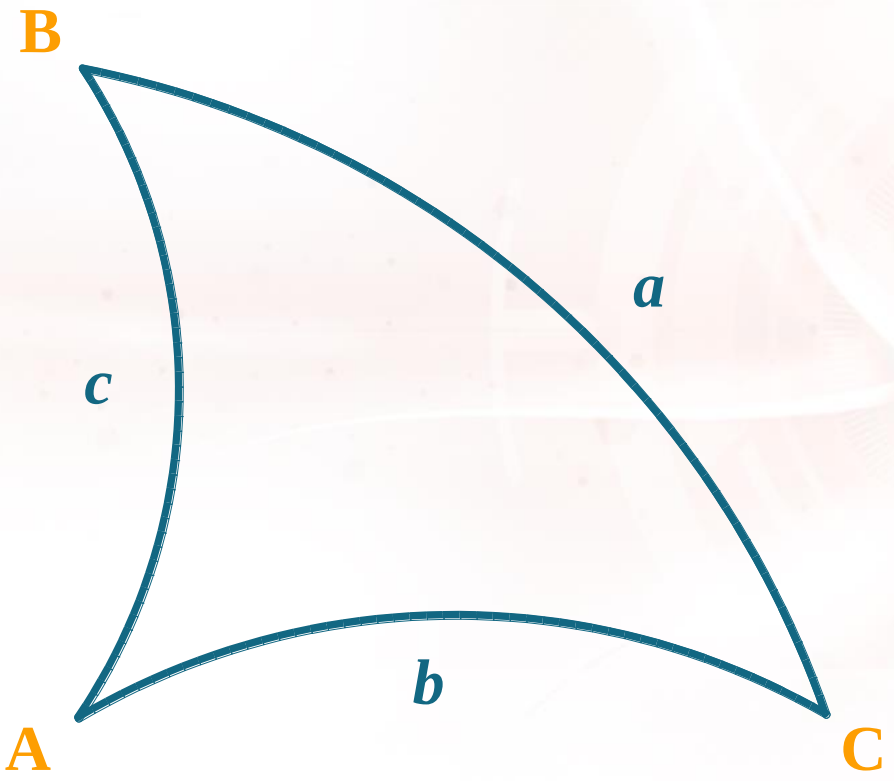
Law of Sines

Law of Cosines

- Sides
- Angles

LAW OF SINES

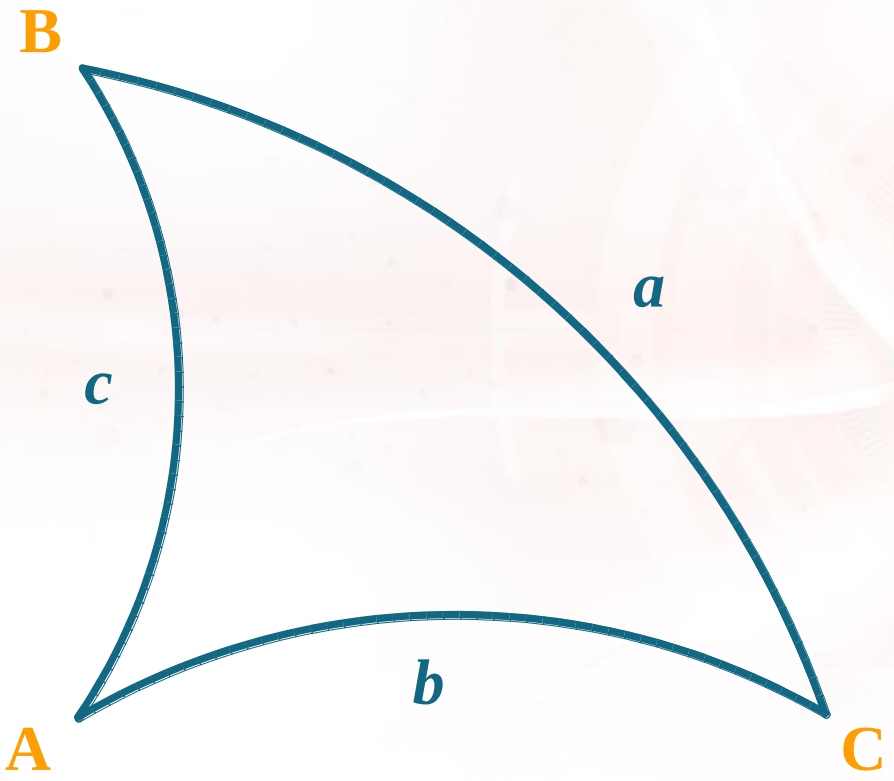
LAW OF SINES



$$\frac{\sin A}{\sin a} = \frac{\sin B}{\sin b} = \frac{\sin C}{\sin c}$$

Solve for **angle** C of the oblique spherical triangle ABC given, $\boldsymbol{a = 80°}$, $\boldsymbol{c = 115°}$ and $\boldsymbol{A = 72°}$.

LAW OF COSINES

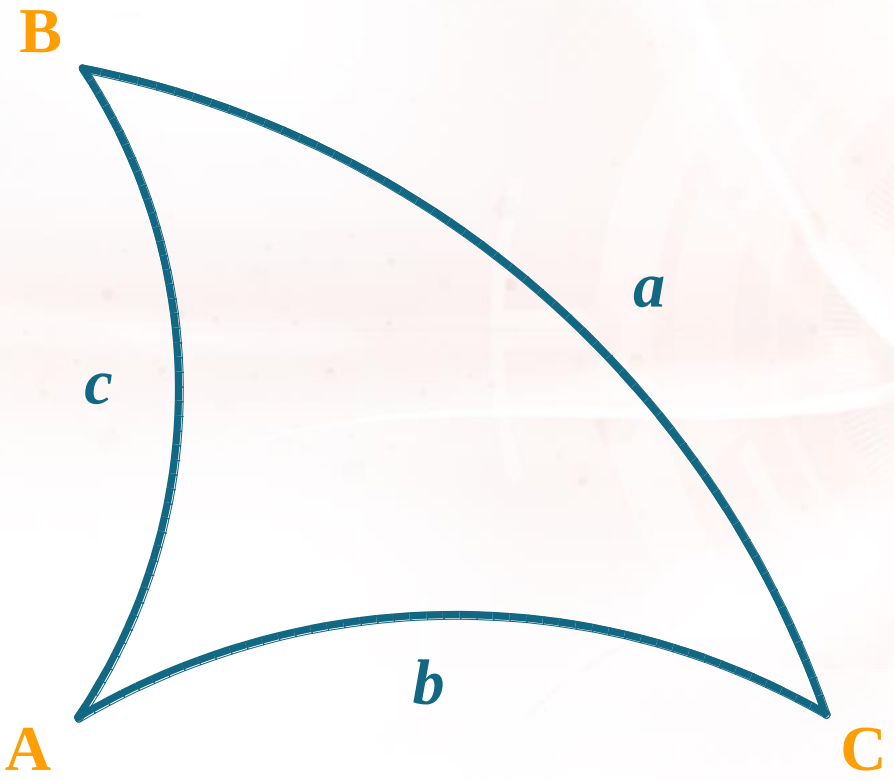


$$\cos a = \cos b \cos c + \sin b \sin c \cos A$$

$$\cos b = \cos a \cos c + \sin a \sin c \cos B$$

$$\cos c = \cos a \cos b + \sin a \sin b \cos C$$

What is the value of the **side c** of a spherical triangle, whose given parts are $\boldsymbol{a = 76°12'}$, $\boldsymbol{b = 51°38'}$ and $\boldsymbol{C = 18°16'}$?



$$\cos A = -\cos B \cos C + \sin B \sin C \cos a$$

$$\cos B = -\cos A \cos C + \sin A \sin C \cos b$$

$$\cos C = -\cos A \cos B + \sin A \sin B \cos c$$

What is the value of the **side a** of a spherical triangle, whose given parts are $\boldsymbol{A = 52°30'}$, $\boldsymbol{B = 70°15'}$ and $\boldsymbol{C = 120°15'}$?

SEATWORK